MD FAHIM FAYSAL

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Rajshahi, Bangladesh

WORK EXPERIENCE

Research Assistant | Bioinformatics Lab (Dry), Department of Statistics, Rajshahi University February, 2024 - Present

- Designed and implemented bioinformatics algorithms, R packages, and web-based platforms for drug and vaccine discovery through integrative analysis of molecular OMICS data, including genomics, transcriptomics, proteomics, and metagenomics.

RemoteIntegrity LLC, Clearwater, Florida, USA | *Software Developer* **January 2025-July 2025**

- Developed a custom CRM system with Node.js, MySQL, and React.js, integrating APIs with platforms like Apploye and YouTube, and built a scalable job application and course management platform attracting over 1000 daily users.

COMPETITIVE PROGRAMMING EXPERIENCES

- **Expert** at **Codeforces Max. Rating 1686**
- Solved 650+ problems in Codeforces
- **Max. Rating 1744** at CodeChef
- Solved 200+ problems in different online judges like Atcoder, Hacherearth, LightOj, LeetCode, etc

EDUCATION

Rajshahi University of Engineering & Technology Feb 2019 - May 2024
Bachelor of Science in Computer Science & Engineering CGPA 3.31/4.00
Adamjee Cantonment College July 2016 - April 2018
Higher Secondary School Certificate GPA 5.00/5.00

PUBLICATIONS IN THE REPUTED JOURNALS

- **Faysal, M. F.**., Noor, T., Shyla, S., Sharma, O., Ahmed, M. F., Ahmed, R., Kabir, M. M. J., & Mollah, M. N. H. Bioinformatics-guided discovery of common genetic underpinnings through which idiopathic pulmonary fibrosis and tuberculosis stimulate each other, and repurposing common therapies. Computers in Biology and Medicine.**[Submitted]**
- **Faysal, M. F.**, Noor, T., Shyla, S., Noor, A., Ahmed, M. F., Sharma, O., Kabir, M. M. J., & Mollah, M. N. H. In-silico discovery of druggable common hub-genes and associated factors through which type-2 diabetes stimulates tuberculosis, and repurposing common drugs through integrated RNA-seq profile analysis. Scientific Reports.**[Submitted]**
- Noor, T.,**Faysal, M. F.**, Ahmed, R., Ahmed, M. F., Sharma, O., Kabir, M. H., Kabir, M. R., Kabir, M. M. J., & Mollah, M. N. H. Decoding shared host key-genes associated with dengue virus and COVID-19 coinfections for diagnosis and targeting shared therapies through integrated RNA-seq profile analysis using bioinformatics approaches. Briefings in Bioinformatics. **[Submitted]**
- Noor, T., **Faysal, M. F.**, Noor, A., Ahmed, M. F., Ahmed, R., Sharma, O., Kabir, M. H., Kabir, M. R., Kabir, M. M. J., & Mollah, M. N. H. Bioinformatics-guided discovery of druggable common host key-genes and associated factors through which type-2 diabetes stimulates dengue virus infection and repurposing common drugs. Scientific Reports. **[Submitted]**

- Ahmed, M. F., Noman, M. A., Ahmed, M. F., Latif, M. A., Pappu, M. a. A., Islam, M. S., Hossain, M. S., Hossen, M. B.,**Faysal, M. F.** F., Hasan, M. M., & Mollah, M. N. H. (2025). In-Silico discovery of Pediatric Acute-Myeloid-Leukemia (pAML) causing druggable molecular signatures highlighting their pathogenetic processes and therapeutic agents through single-cell RNA-Seq profile analysis. *PLOS ONE*, 20(10), e0335410. https://doi.org/10.1371/journal.pone.0335410
- Pappu, M.A.A.; Alamin, M.; Noman, M.A.; Sultana, M.H.; Ahmed, M.F.; Hossain, M.S.; Latif, M.A.; **Faysal, M. F.**; Azad, A.; Alyami, S.A.; et al. Multi-Transcriptome-Informed Network Pharmacology Reveals Novel Biomarkers and Therapeutic Candidates for Parkinson's Disease.https://doi.org/10.3390/genes16121459
- Ahmed, M. F., **Faysal, M. F.**, Hossen, M. B., Sarker, A., Noman, M. A., Ahmed, M. F., Murad, M. M. H., Begum, A. A., & Mollah, M. N. H. Bioinformatics-based discovery of shared pathogenetic mechanisms by which type-2 diabetes stimulates hepatitis-B-virus infection, and repurposing common drugs. **[under review]**
- Uddin, B. A., Sarker, A. A., **Faysal, M. F.**, & Mollah, M. N. H. . Unveiling drug candidates against human metapneumovirus infection by targeting the RNA-dependent RNA polymerase (RdRp) through bioinformatics analysis.**[under review]**
- Ahmed, M.F., Sharma, O., Noor, T., **Faysal, M. F.**, Latif, M. A., ..., & Mollah, M. N. H. . Disclosing shared key molecular signatures and their mechanisms in triple-negative breast cancer (TNBC) and type-2 diabetes (T2D) via machine learning based filtering for repurposing unique therapies. *Briefings in Bioinformatics*.**[under review]**
- Ahmed, R., Antu, U.H., Noor, T., **Faysal, M.F.**, Akter, M.T., Nesa,M., & Mollah, M. N. H. .Bioinformatics-guided discovery of shared molecular signatures and their functions through which type-2 diabetes (T2D) stimulate myocardial infarction (MI), and repurposing candidate drugs effective against both diseases. *Computational Biology and Chemistry*.**[under review]**
- Hussain, M. R., Ahsan, M. A. M., Islam, M. S., Ali, M., Ahmed, M. F., Noman, M. A., **Faysal, M. F.**, Hossain, M. S., Sipa, U. T. A., Kabir, M. H., & Mollah, M. N. H. . In-silico detection of genetic variants associated with non-alcoholic fatty liver disease (NAFLD) through meta-analysis of genome-wide association studies for diagnosis and therapies. *PLOS ONE*. **[under review]**
- Mahmud, S., Ajadee, A., Ali, M. A., Rana, M. M., Ahmmed, R., Noor, T., **Faysal, M. F.**, Kabir, M. H., & Mollah, M. N. H. . Disclosing pathogenetic processes of monkeypox (MPOX) and drug repurposing through host-transcriptomics analysis. *PLOS Neglected Tropical Diseases*.**[under review]**

PUBLICATIONS IN THE CONFERENCE PROCEEDINGS

- **Faysal, M. F.**, Afroge, S., & Noor, T.. (2024, November). In-Silico Identification of Shared Hub-Genes Between Idiopathic Pulmonary Fibrosis and Tuberculosis Diseases & Drug Repurposing. In 2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON) (pp. 67-72). IEEE.
- Noor, T.,**Faysal, M. F.**, Sadad, N., & Mondal, M. N. I. (2024, December). Discovery of SARS-COV-2 and Dengue-Virus Co-Infections Causing Shared Host Key-Genes using Bioinformatics and Machine Learning Approaches for Diagnosis and Therapies. In Proceedings of the 8th International Conference on the Role of Statistics and Data Science in 4IR (ICRSDS4IR). Rajshahi, Bangladesh.

TECHNICAL SKILLS

- **Programming & Scripting** C, C++, Java, JavaScript (Node.js, React.js, Angular.js), Python, R, HTML, CSS, SQL (MySQL, PostgreSQL)
- **Machine Learning/Deep Learning:** NumPy, Pandas, Matplotlib, OpenCV, Scikit-learn,Random Forest
- **Bioinformatics & Computational Biology:** LIMMA, DESeq2, WCGNA, PPI Network Analysis, Data Visualization (ggplot2, ComplexHeatmap, Plotly), Biomarker Discovery, Molecular Docking (AutoDock Vina), ADMET Analysis (SwissADME, pkCSM, ADMETlab), Drug Repurposing Pipelines
- **Cheminformatics & Structural Biology:** RDKit, Open Babel, PyMOL, Discovery Studio Visualizer.

AWARDS

- Leadership Award Class Representative 2019-2021